

Hyperplane

Point and vector

- A point in n-dimensional space is given by an n-tuple, e.g., $P = (p_i)$
 - A point denotes an absolute position in space
- A vector represents magnitude and direction in space, and is given by an n-tuple.
 - Vectors do not have a fixed position in place, but can be located at any initial base point P.
- The vector from point P to Q is given by $v = Q - P = (q_i - p_i)$
- Vector addition: $v \pm w = (v_i \pm w_i)$
- The length of a vector v : $|v| = \sqrt{\sum_{i=1}^n v_i^2}$
- http://geomalgorithms.com/points_and_vectors.html

Normal vector

- A normal vector is a vector perpendicular to another object (e.g., a plane).
- A unit normal vector is a normal vector of length one.
If N is a normal vector, the unit normal vector is $N/|N|$, where $|N|$ is the length of N .

The equation for a hyperplane

- A 3-D plane determined by normal vector $N=(A, B, C)$ and point $Q=(x_0, y_0, z_0)$ is:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

which can be written as

$$Ax + By + Cz + D = 0, \text{ where } D = -A x_0 - B y_0 - C z_0$$

- Hyperplane: $w \cdot x + d = 0$,
where w is a normal vector, x is a point on the hyperplane
 - It separates the space into two half-spaces:
 $w \cdot x + d > 0$ and $w \cdot x + d < 0$

The distance from a point to a plane

- Given a plane $Ax + By + Cz + D = 0$, and a point $P=(x_1, y_1, z_1)$, the distance from P to the plane is:

$$d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

See http://mathinsight.org/distance_point_plane

- Distance from a point x to a hyperplane $wx + d = 0$ is:

$$|wx + d| / ||w||$$

Distance between two parallel planes

- Two planes $A_1x + B_1y + C_1z + D_1 = 0$ and $A_2x + B_2y + C_2z + D_2 = 0$ are parallel if $A_1 = k A_2$, $B_1 = k B_2$ and $C_1 = k C_2$
- The distance between $Ax + By + Cz + D_1 = 0$ and $Ax + By + Cz + D_2 = 0$ is equal to the distance from a point (x_1, y_1, z_1) on the first plane to the second plane:

$$\frac{|Ax_1 + By_1 + Cz_1 + D_2|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|D_2 - D_1|}{\sqrt{A^2 + B^2 + C^2}}$$

- The distance between two hyperplanes $wx + d_1 = 0$ and $wx + d_2 = 0$ is

$$\frac{|d_2 - d_1|}{\|w\|}$$